

Submit your solution on Canvas.

Do not discuss these problems with other students. You should solve these problems on your own.

Problem 1 (25 points). Consider a cache that uses the Least Recently Used (LRU) page replacement policy. Suppose that our computer program requests completely random pages: Every second it picks a random number r from 1 to n and requests page number r . Find the competitive ratio of the LRU policy:

$$\text{expected competitive ratio} = \mathbf{E} \left[\frac{\# \text{ LRU page misses}}{\# \text{ page misses in OPT}} \right].$$

The total number of pages is n ; the size of the cache is k pages; and the total number of requests is T .

A. Find the expected competitive ratio of LRU assuming that $T \gg n$ and $n \gg k$.

B. Find the expected competitive ratio of LRU assuming that $T \gg n$ and $n = k + 1$.

In the both parts, you can assume that k , n , and T are sufficiently large.

Problem 2 (25 points). Design FPT algorithms for the following problems and prove that they are correct.

1. Finding a cycle of length exactly k in a graph G .
2. Finding a cycle of length from k to $2k$ in a graph G .
3. Finding a cycle of length at least k in a graph G .

Hint. Prove the following claim: If a graph G contains a cycle of length at least $2k$, then after contracting an arbitrary edge, it still contains a cycle of length at least k .

Problem 3 (25 points). Consider the following linear program (LP).

$$\min 5x_1 + 3x_2 + 2x_3$$

subject to

$$x_1 + x_2 + x_3 \geq 6$$

$$x_1 + x_2 \geq 5$$

$$x_1 - x_3 \geq 1$$

$$x_1 - x_2 \geq 1$$

$$x_1, x_2, x_3 \geq 0$$

1. Write the dual LP for the LP above.
2. Solve the primal and dual LPs: Find the optimal solutions x^* and y^* and then compare the LP values for x^* and y^* .
3. Verify that the complementary slackness conditions hold for x^* and y^* .